Control Panel

Forecast Model

Random Forest

Focus Machine

Auto: Highest Risk

History Window

2024-01-01 2024-08-27

Ambient Temperature Filter (F)

45.00 98.00

30-Day Alert Threshold

0.45

🏭 Industrial AI Factory Control Room

Digital twin operations dashboard for fleet health, failure forecasting, and maintenance prioritization.

Model: Random Forest | **Focus Machine:** MCH-022 | **Window:** 2024-01-01 → 2024-08-27

Machines in View	Critical	High Priority	Avg Health	Avg 30D Risk
60	4	15	62.9	37.3%

Critical High Watch Normal

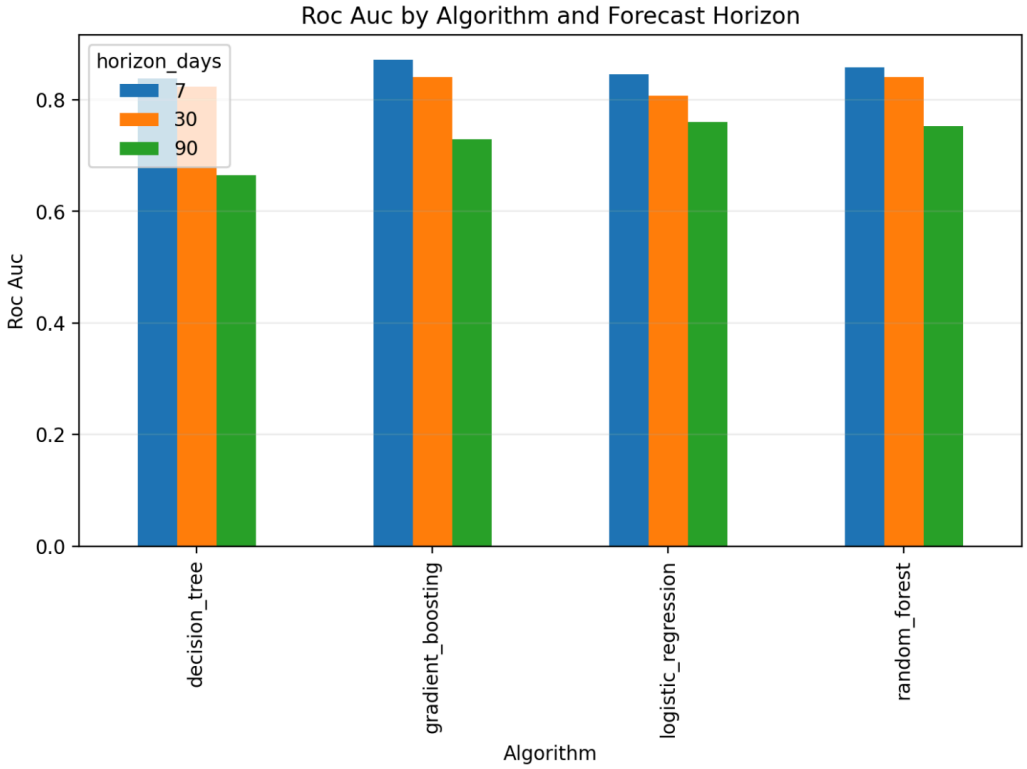
Control Room Factory Map Machine Digital Twin Model Intelligence

Model Performance Intelligence

target	horizon_days	model	accuracy	precision	recall	f1_score	roc_auc	pr_auc	threshold
failure_next_7d	7	gradient_boosting	0.8278	0.4377	0.708	0.541	0.8725	0.5744	0.24
failure_next_7d	7	logistic_regression	0.8122	0.4125	0.7313	0.5275	0.8464	0.4535	0.63
failure_next_7d	7	random_forest	0.8581	0.505	0.5194	0.5121	0.8584	0.5488	0.67
failure_next_7d	7	decision_tree	0.8656	0.5349	0.4755	0.5034	0.8391	0.5042	0.76
failure_next_30d	30	random_forest	0.7607	0.6456	0.8251	0.7244	0.8407	0.7619	0.59
failure_next_30d	30	gradient_boosting	0.7333	0.6047	0.8669	0.7125	0.8408	0.7569	0.48
failure_next_30d	30	logistic_regression	0.7426	0.6212	0.8319	0.7113	0.807	0.6655	0.47
failure_next_30d	30	decision_tree	0.7211	0.5911	0.8698	0.7039	0.8236	0.6954	0.42
failure_next_90d	90	logistic_regression	0.6319	0.5324	0.8778	0.6628	0.7606	0.6954	0.48
failure_next_90d	90	random_forest	0.6348	0.5355	0.8616	0.6605	0.7526	0.6455	0.78

Performance Metric

roc_auc



Built specifically for the industrial_time_series_dataset_v2 schema and the associated trained model artifacts.